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Project #036: Stroke Alert to Triage, Improving Ambulatory Stroke Patient Throughput

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Stroke Alert to Triage: Improving Ambulatory Stroke Patient Throughput

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Abstract

Decreased time to intervention for patients experiencing stroke is associated with improved outcomes including decreased morbidity and mortality.

Goal: To decrease stroke patient throughput and door to TNK time

Problem Description: Throughput times for Stroke patients arriving through the main ED/ambulatory entrance were not meeting benchmark metrics. Prior to intervention patients arriving through the ambulatory entrance with Stroke-like symptoms were moved to the resuscitation room for provider evaluation, glucose check, IV placement and CT order entry. This process was not efficient enough to meet Door to Provider, Door to CT & Door to Intervention time goals. The process changed to bring the provider to the patient in triage upon recognition of Stroke-like symptoms. This allowed for quick provider evaluation and transport of the patient to CT directly from triage. This resulted in dramatically decreased Stroke throughput times.

Do

Intervention: The ambulatory Stroke throughput process was changed for ambulatory patients. Upon identification of Stroke symptoms, the Triage nurse pages overhead "Stroke alert to triage". The ED provider then goes to triage to evaluate for Stroke activation. The patient is then taken directly to CT from triage.

Measures: Stroke throughput times were tracked and compared to our goals:

- Door to Provider time goal is under 5 minutes.
- Door to FAST under 15 minutes.
- Door to CT time goal under 10 minutes.
- Door to TNK time under 45 minutes with a stretch goal of under 30 minutes.

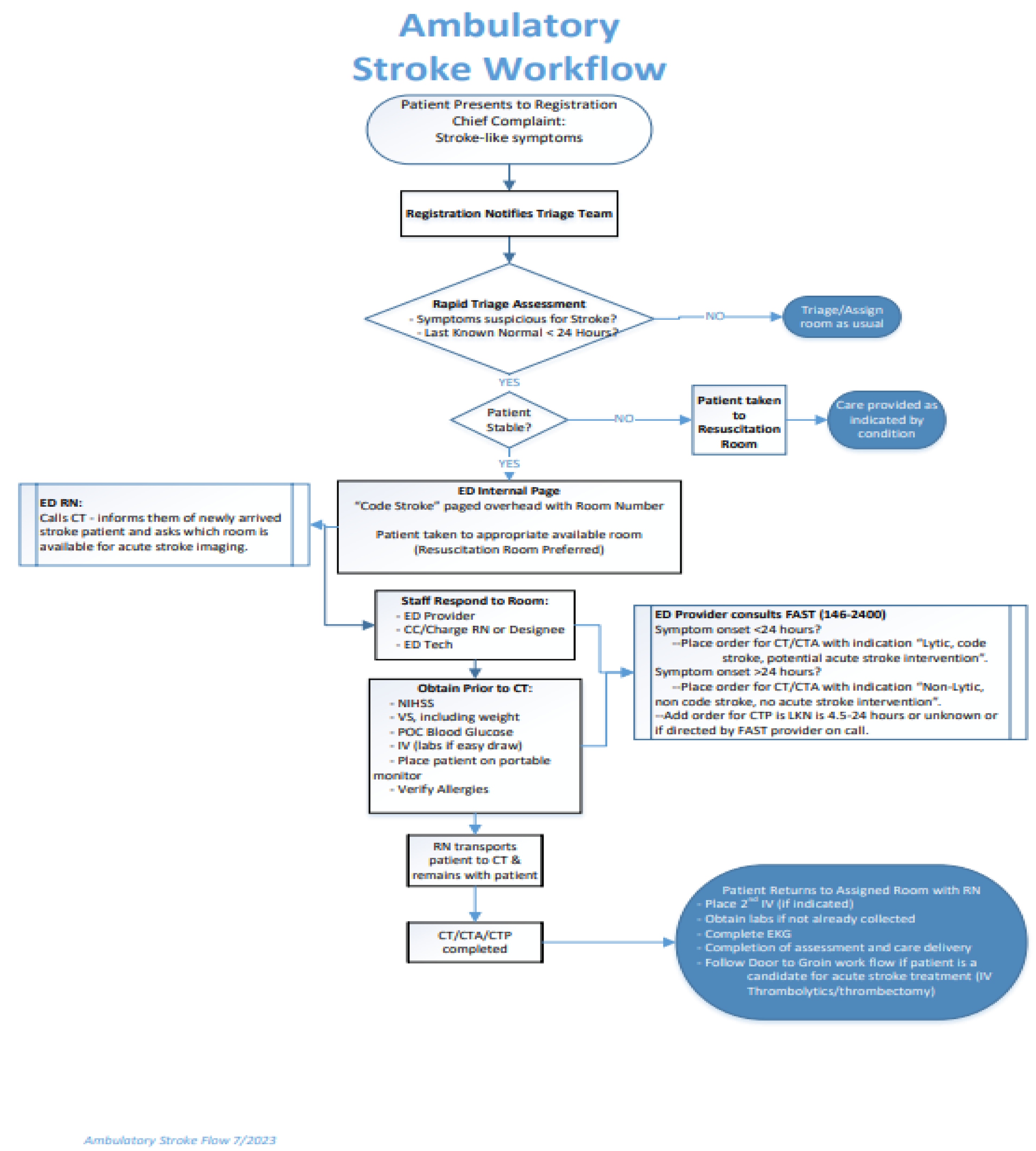
Check

The ED providers and Nursing team were educated on the new process for ambulatory Stroke patients, go-live date was set. Leaders were present in the early weeks of the process to ensure that this workflow became hardwired.

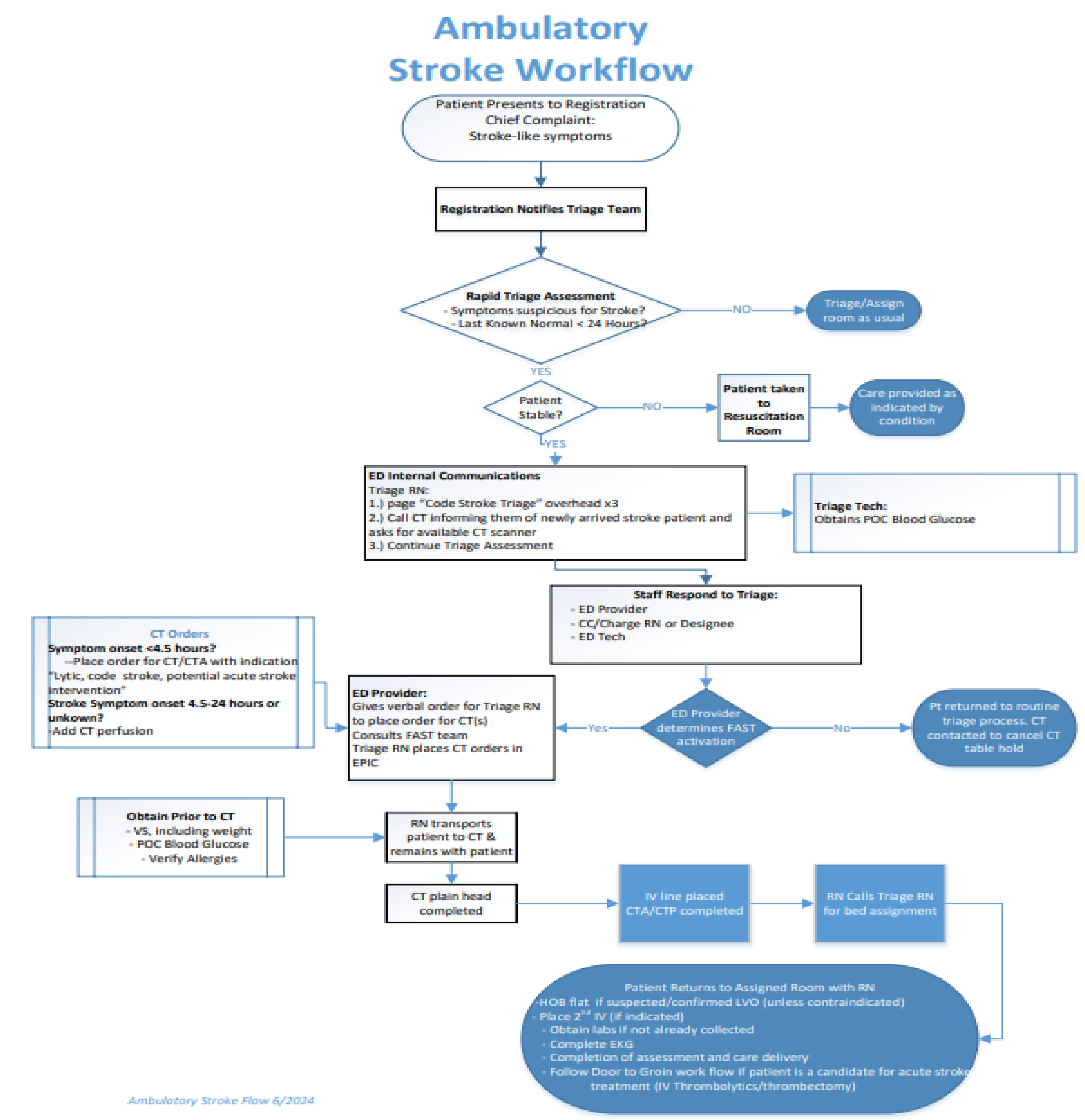
Upon implementation we identified the following opportunities to further streamline our new workflow:

- Paging overhead "Stroke alert to triage" x1 was not successful in notifying the providers this was changed to paging x3 to ensure the providers were able to hear the page.
- Patients were arriving to CT before the order had crossed over to radiology. We added the step of the RN placing the verbal order from the provider in triage. This change allowed for the 2-5 minutes for order cross over to occur while the patient was being transported to CT and set on the CT table.

Pre-Intervention workflow



Post- Intervention Workflow



Stroke Throughput Improvements

2024 Stroke Throughput Times

Metric	Goal	1	2	3	4	5	6	7	8	9
Door to MD	<5 minutes	0:09	0:22	0:10	0:01	0:03	0:08	0:08	0:01	0:02
Door to FAST	<15 minutes	0:13	0:28	0:42	0:01	0:03	0:09	0:07	0:07	0:10
Door to CT	<15 minutes	0:25	0:30	0:33	0:06	0:15	0:18	0:14	0:07	0:09
CT Read	<10 minutes	0:09	0:17	0:10	0:13	0:13	0:09	0:08	0:18	0:15
Door to CT Read	<25 minutes	0:34	0:48	0:43	0:21	0:21	0:26	0:22	0:23	0:24
Door to Needle	<45 minutes	1:05	1:14	1:05	0:45	0:59	1:03	0:43	0:32	0:37

Intervention

2025 TNK Throughput Times

Metric	Goal	1	2	3	4	5	6	7
Door to MD	<5 minutes		0:00	0:01	0:03	0:04	0:02	0:08
Door to FAST	<15 minutes		0:10	0:09	0:15	0:17	0:02	0:10
Door to CT	<15 minutes		0:07	0:08	0:10	0:15	0:07	0:17
CT Read	<10 minutes		0:16	0:21	0:12	0:08	0:11	0:12
Door to CT Read	<25 minutes		0:23	0:31	0:18	0:25	0:20	0:29
Door to Needle	<45 minutes		0:32	0:59	0:51	0:58	0:38	0:57

Act/Discussion

Summary:

- By bringing the provider to the patient at point of entry Stroke throughput metrics improved.
- The previous process of placing a patient in a room prior to initiating stroke workflows contributed to extended throughput times.
- This was a budget- neutral initiative as no additional staffing, equipment or construction was needed to implement this process.
- Moving Patients to a room in the ED prior to beginning evaluation and intervention increased time to Thrombolytics and/or Thrombectomy.

Conclusions:

- This Stroke process improved our response to this time-sensitive emergency.

References/Acknowledgements

The West Bloomfield ED Nursing and Provider teams for adopting this new workflow.